

Other issues such as concurrency, cost, compliance control, robot locomotion (particularly for legged, amphibious and flying robots), human-computer interactions and cognitive computer vision are also facing a great

deal of scrutiny with considerable research going on in each area.

A high level of specialisation is required to work on any particular aspect of robotics. Hence there are not too many options for taking up

robotics as a degree itself, especially at the undergraduate level. Robotics still remains a popular course as an elective subject for engineering students.

Opportunities in India

Innovation coupled with consolidated research and development in robot technology has catapulted India's scientific position to a level that equals other advanced countries in recent years. It is, however, a known fact that robots are not widely used by Indian companies. Out of 600 to 700 robots widely used in India, most are used in engineering institutes and research organisations (see Tables I and II).

Robotics education is taking shape and many universities are introducing courses in robotics and automation at the postgraduate level as main or elective subjects. Seminars, workshops and conferences are also conducted by universities and institutes at regular intervals to project the recent developments in robotics technologies at national and international levels. These activities have inspired a notable number of under-graduate students to pursue higher studies in the field of robotics and automation in India and abroad. Students' participation in various robotics contests, meets and expos at international level has also increased marginally in

TABLE I
Engineering Institutes Offering Courses in Robotics in India

Institute	Course offered
Centre for Robotics and Mechatronics at IIT Kanpur	M.Tech in robotics engineering
University of Hyderabad	M.Tech in artificial intelligence and robotics
University College of Engineering, under the Osmania University, Hyderabad	ME/M.Tech in automaton and robotics
MS University, Baroda	Master's programme with specialisation in automatic control and robotics
Institute of Technology, Banaras Hindu University	M.Tech in production engineering/industrial management engineering with robotics and automation as electives
The Birla Institute of Technology and Science (BITS), Pilani	ME in computer science with robotics as a subject
Department of Engineering Design, IIT Madras	Dual degree programme in engineering design and automotive engineering
Sri Sathya Sai Institute of Higher Learning, Prasanthinilayam, Andhra Pradesh	M.Tech in computer science with specialisation in artificial intelligence
The PSG College of Technology, Coimbatore	M.Tech in mechanical engineering with robotics as an elective subject
SRM University, Chennai	M.Tech in robotics
Cochin University of Science and Technology (CUSAT), Kerala	M.Sc in electronic science with specialisation in artificial intelligence and robotics
Defence Institute of Advanced Technology (DIAT), Pune	M.Tech in automotive engineering

TABLE II
Robotics Research Centres in India

Governing body	Centres	Applications
IIT Kanpur	Centre for Robotics & Mechatronics	Surgical robots and humanoid robots
NIT Rourkela	Centre for Industrial Electronics & Robotics	Industrial robots
Sathyabama University	Centre for Robotics	Robotics design
IIIT, Hyderabad	Robotics Research Lab	Robotics vision
IIT Madras	Department of Design Engineering	Underwater and medical robotics
Defence Research Development Organization (DRDO)	Centre for Artificial Intelligence & Robotics, Bangalore CVRDE, Chennai VRDE, Ahmednagar R & D E (E), Pune ADE, Bengaluru NSTL, Vizag NPOL, Kochi	Mini/micro robots Unmanned ground vehicles Autonomous surveillance vehicles Remotely operated underwater vehicles Unmanned aerial vehicles Autonomous underwater vehicles Underwater surveillance robots
Council of Scientific & Industrial Research (CSIR)	CSIO, Chandigarh CEERI, Pillani	Automation Industrial automation
Department of Atomic Energy (DAE)	BARC Mumbai	Laser-based robot for mapping
Ministry of Earth Sciences	NIOT, Chennai	Remote-operated boats